Mill Startup Prompting User's Guide

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This document details the use of a function for identifying unsatisfied mill permissives.

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1 Preface

1.1 Intended Audience

This manual is written for operators, computer staff, and system managers. Sections 1 to 3 provide general information for any reader. Operators should read the section called "Operator Interface". The section on "Maintenance" provides details to the system manager or "database administrator" who will configure and maintain MSUP. The Appendix outlines steps for installation.

2 Description

The GE Mill Startup Prompting (MSUP) function provides the operator with an automated menu-driven approach to identifying unsatisfied mill permissives. The MSUP function is organized as a permissive "tree", with major subsystems at the highest levels of the tree, branching out into minor subsystems and finally individual mill permissives at the lowest levels of the tree.

At each level of the tree the operator sees the status of all subsystems which comprise that level. At the highest level, the entire mill is either "READY" or "NOT_READY", based on the state of all of its subsystems (e.g. roughing mill "READY", finishing mill "READY", coiler "READY", etc). The user can interactively traverse the tree looking at subsystem states, or immediately locate permissives which are keeping the mill from being "READY". Each time a particular node is traversed, the current state of any corresponding permissive is determined.

Each node in the tree has a short description which is displayed to the operator as he traverses the tree. This description may also describe corrective action to take if the permissive is unsatisfied.

3 Requirements

Control System Freeway

The MSUP software obtains information about mill permissives by issuing "directed messages" on the Control System Freeway (CSF). The associated CSF hardware and software must be present for MSUP to function.

Distributed Micro Controller

Mill permissives must be resident in Distributed Micro Controllers (DMC). The DMC must be running revision (3.4) or later software.

Fault or Permissive Tree

GE Application Engineers, GE Field Service (ISE), or customers can be responsible for building a permissive tree.

VAX/VMS Operating System

This product will operate on VAX/VMS version 4.4 or later operating systems.

DEC Forms Management System

This function requires the DEC FMS (Forms Management System) layered product for VMS, utilizing any DEC VT-type terminal.

4 The MSUP Fault Tree

4.1 Description

The fault tree describes the permissive logic used to determine if the mill is ready or not ready. At the highest level are Major Subsystems, at the next level Minor Subsystems, and then one or more levels of permissives. An example follows:

```
Major Subsystem
- - Minor Subsystem
- - - - Permissive Level 1
- - - - - - Permissive Level 2
- - - - - - - - Permissive Level 3
- - - - - - - - - - Permissive Level 4

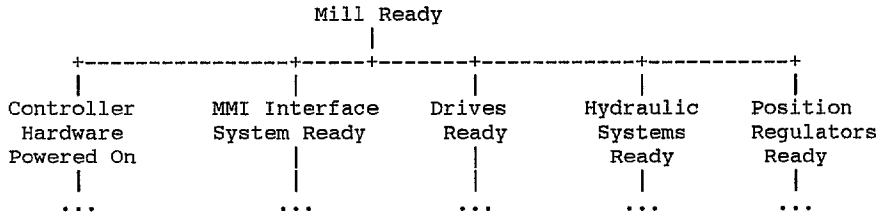
FM Level I Automation Online
   HCC/AGC DMCs Online
      Stand F6 HCC/AGC DMC Online
         Power Supply On
            Panel Breaker Closed
            AC Distribution Breaker Closed
         Module I/O Online
         Remote I/O Station Online
            Power Supply On
               Panel Breaker Closed
               AC Distribution Breaker Closed
            I/O Fuses Intact
               I/O Fuse #1 Intact
               I/O Fuse #2 Intact
                     .
                     .
                     .
               I/O Fuse #n Intact
```

This example illustrates only a very small part of a system fault tree. The reader can appreciate that the full definition of the system tree is not trivial. For this reason, this document describes a general purpose tool for traversing an application specific fault tree.

5 Operator Interface

The operator interfaces with the MSUP fault tree via the "MSUP" routine. The MSUP program generates displays depicting the fault tree on the Major and Minor Subsystem levels, as well as on the permissives' level. On each display the operator is presented with a list of subsystems in the preferred order of startup. The status (READY, NOT READY, UNKNOWN, etc.) of each subsystem is listed alongside of its description. The operator can traverse the fault tree by selecting one of the branches from the menu, thereby working down the tree until the desired permissive is reached. The operator also has the ability to "pop" back up to a "parent" node, or return to the highest level of the fault tree. The operator is not permitted to alter the database in any way. The following diagram depicts the top level of a sample tree. Following this diagram is a figure showing what the operator will see when entering the MSUP program.

SAMPLE PERMISSIVE TREE



SAMPLE MSUP SCREEN

```
 MSUP                                                  08:34:51  25-APR-89
+--------------+-----------------------------------------------------+
| - Search     |    Top of fault tree                                |
| - Up         |          Current Node = NOT_READY                   |
| - Top        |              DMC Name =                             |
| - Refresh    |         Variable Name =                             |
| - Exit       |                                                     |
+--------------+-----------------------------------------------------+
|   Status          Branches from current node                       |
|                                                                    |
| _ READY       -  Controller Hardware Powered On                    |
| _ READY       -  MMI Interface System Ready                        |
| _ NOT_READY   -  Drives Ready                                      |
| _ READY       -  Hydraulic Systems Ready                           |
| _ READY       -  Position Regulators Ready                         |
|                                                                    |
|                                                                    |
|                                                                    |
|                                                                    |
+--------------------------------------------------------------------+
(Position cursor using TAB/BKSP, UP/DOWN arrow, or 1st letter of
 COMMAND)  Press RETURN or ENTER to select that line.  Press PF2
 twice for HELP.
```

The MSUP screen is functionally divided into four sections. Each of these areas of the screen are discussed in the following sub-sections.

5.1 MSUP Screen - Current Node

The upper right-hand boxed area shows the operator the status of the current "node" in the permissive tree. When the operator first enters MSUP, this node is the top of the tree, i.e. the mill ready node. The "current node" shows the logical state of this node (e.g. READY, NOT_READY). If the current node is a leaf node, then the name of the DMC and the associated variable name for this permissive is also displayed in the upper right hand area of the MSUP screen.

5.2 MSUP Screen - Branches

The lower boxed area of the MSUP screen shows all branches beneath the current node, the status of each of these branches, and a short description for each branch. If the current node is at a "bottom level" or "leaf node" of the tree, there will be no branches.

5.3 MSUP Status Values

Possible status values are listed below. The return status values of MSUP may be due to a successful status request from a DMC, an error in the MSUP input database (permissive tree), or CSF channel problems. Please refer to Section 6 for a description of each status value.

5.3.1 Normal Status Values

- READY - Node in correct state.
- NOT_READY - Node not in correct state. Permissive not satisfied.

5.3.2 Permissive Tree Errors

Permissive tree errors are generated when MSUP encounters errors in the input database (MSUP.DATABASE). Contact the "MSUP database administrator" for your site to report errors in the permissive tree.

- NOT_IN_DMC
- NOT_A_LOGICAL
- BAD_LEAF_NODE
- BAD_BYPASS
- STN_NOT_FOUND

5.3.3 CSF Errors

A number of errors can be generated when MSUP is unable to successfully communicate over the Control System Freeway. For the majority of the following errors, use of the REFRESH command can result in the error being replaced by a successful status return value.

- DMH_STAT_FAIL
- BMP_SEND_ERR
- BMP_RCV_FAIL
- BAD_PACKET_ID
- DMC_Q_IS_FULL
- TIME_OUT
- BAD_DMC_STATE - DMC is not online.
- NOT_ACKED
- CHANNEL_FAIL
- UNKNOWN

5.4 MSUP - Commands

The upper-left hand boxed area of the MSUP operations screen displays commands available to the operator. These commands may be accessed by pressing the up or down arrow keys, the tab and backspace keys, or pressing the letter on the keyboard that corresponds to the first letter of the command (i.e. "S" means Search). The letter "B" for Branch may also be pressed to position the operator next to the first branch under the current node. Once the operator is positioned next to the desired command, press the RETURN key and the command will be executed. If the cursor is next to a valid branch under the current node, the action of pressing the RETURN key will move this branch into the "current node" section of the screen, and any/all branches beneath this selected node will now be displayed in the lower half of the screen.

SEARCH Command

The SEARCH command searches the current branch of the mill startup fault tree, and the current node pointer is reset to the location of the first unsatisfied permissive found in the sub-tree. If no unsatisfied permissive is found, the current node is not reset, and an informative message is displayed to the operator.

When a desired permissive is reached, the description of this node provides information about the location and action to be taken by the support personnel to effect repairs. This information is stored in the description field, and is therefore available to the operator without having to select each individual permissive.

A word of caution is needed for the SEARCH command. To improve efficiency, the directed messages to each DMC are buffered, and actually sent out to the DMC when a buffer is full, or when an entire sub-tree has been searched. The permissive found by the SEARCH command might therefore correspond to a variable which is not logically "first" in a tree. At any rate, this command will always find an unsatisfied permissive if any is present in the sub-tree beneath the current node. In this way, the operator may select which sub-tree to search before pressing the SEARCH command.

UP Command

The UP Command resets the current node to the "parent" node above it. This command makes it possible for the operator to traverse the permissive tree in both directions : down to the "leaf" nodes, or up to the top of the tree. Each UP command positions the user one level higher than the current screen. The node which was listed as the "current node" will now be seen as one of the branches from the new screen. This command will have no effect when the user is at the top of the permissive tree, and an alarm will be displayed along the bottom of the screen indicating that the user is already at the top of the tree.

TOP Command

The TOP command resets the current node to the highest point in the permissive tree, often referred to as the "Mill Ready ?" node. This provides a quick and efficient means for the user to traverse sub-trees and then return to the top of the tree. If the user is already at the top of the permissive tree, this command will have no effect, and a message will be displayed along the bottom of the screen indicating that the user is already at the top of the tree.

REFRESH Command

The REFRESH Command allows the operator to obtain a new set of status values for the current node/branches displayed on the screen. MSUP communicates with each DMC required to obtain a new set of status values. This command is useful when communication errors preclude the operator from obtaining actual DMC variable status values by issuing a "re-try" for all the values listed on the screen. Bear in mind that the MSUP screen status values are not "live"; to see if a problem has been corrected, the user must use this REFRESH command.

EXIT Command

Placing the cursor next to this command and pressing RETURN allows the user to exit the MSUP utility.

5.4.1 MSUP - HELP Screens

The final area to be described on the MSUP operations screen is outside of the boxed area, at the bottom of the screen. Two lines of "help" text are always present, followed by a third blank line. When an operator presses the PF2 key (function key #2) on the VT terminal keypad, a single line of help text is displayed, describing the command or line next to the current cursor position. If the operator presses the PF2 key a second time, a full page of HELP text will appear, documenting most of the MSUP return status values. This HELP screen is re-created in the following figure :

MSUP - HELP SCREEN

```
Mill Startup Prompting -  The return status values of MSUP may be
                          due to a successful status request from
                          a DMC, an error in the MSUP input database,
                          or CSF channel problems.  Channel or status
                          values might therefore change if a REFRESH
    RETURN STATUS         request is issued.
+-------------------------------------------------------------------------+
| CHANNEL_FAIL  - Unable to open a channel to the DMC due to other users  |
| BAD_DMC_STATE - DMC not on-line (e.g. DEAD)                             |
| BAD_LEAF_NODE - A leaf node was found without a permissive.             |
| BAD_PACKET_ID - The sequence # sent to DMC does not match one returned  |
| DMC_Q_IS_FULL - Message queue for desired DMC is full.                  |
| BMP_RCV_FAIL  - Timeout receiving request. DMH subprocess needs reset?  |
| BMP_SEND_ERR  - Likely timeout sending request.  Please try again.      |
| DMH_STAT_FAIL - Message sent out, but DMC did not respond.              |
| NOT_ACKED     - DMC did not acknowledge the message sent to it.         |
| NOT_A_LOGICAL - Requested symbol is in DMC, but not a logical value.    |
| NOT_IN_DMC    - DMC is online and healthy : symbol not found in DMC.    |
| NOT_READY     - DMC recognizes logical, and reports status as FALSE.    |
| READY         - DMC recognizes logical, and reports status as TRUE.     |
| STN_NOT_FOUND - DMC station not included in Station Configuration File  |
+-------------------------------------------------------------------------+
```

6 Maintenance

The following sections are written for the "database administrator" or system manager.

6.1 The Database Administrator

The "database administrator" for the MSUP function is the individual(s) responsible for the creation and maintenance of the MSUP fault tree. This person must be able to add, modify, or delete nodes in the fault tree. It is these nodes which the operator traverses when attempting to find an unsatisfied permissive (mill is not ready for startup).

The MSUP design allows the database administrator to visualize the tree structure being constructed or modified. Permissive trees can be constructed out of multiple input files, created using any file editor, which are then "preprocessed" to create the Mill Startup Prompting database. To visualize the multiple levels of the fault tree, a system of indentation is used, which, when coupled with the include file capability, allows specification of files for sub-branches of the tree. Include files may in turn specify other include files, allowing the otherwise unmanageably large database to be broken down into manageable files.

The MSUP permissive tree (database) has been designed to allow the specification of "reverse logic" permissives. Any permissive symbol name beginning with a Tilde character (~) is interpreted by MSUP to indicate a permissive whose normal state is FALSE. MSUP strips away the (~) character before interrogating the DMC for this symbol's status. If the return status of any symbol beginning with a Tilde is FALSE, MSUP returns READY as the actual status. If the return status is TRUE, then for a reverse-logic node, MSUP flags this node as being unsatisfied (NOT_READY).

The MSUP database provides a special node known as an "override" or "bypass" node. These special override nodes require the MSUP program to immediately issue a directed message to obtain the value of the permissive specified by these nodes. If this permissive is TRUE, then the sub-tree beneath these override nodes is skipped ; the sub-tree is "overridden" and a status of READY is returned for that node. If the override permissive is FALSE, then the sub-tree is traversed in the normal way. Note that a sub-tree is required beneath these nodes. The reverse logic capabilities discussed above are also allowed for override nodes. An override node is specified in the MSUP database by entering a pound sign (#) before the dmc symbol name. The reverse-logic Tilde character (~) and the override node pound sign (#) are order independent in the msup database (i.e. may be character 1 or 2 when BOTH are specified).

The following section describes the utility the database administrator utilizes to create the MSUP database.

6.2 CONVERT TREE Utility

The CONVERT TREE utility is provided to the MSUP database administrator as a tool for creating the MSUP fault tree. The description of this utility follows.

6.2.1 Creating the input file

This preprocessor allows (encourages!) the specification of separate include files to define sub-tree structures. Any input file may contain comment lines on any line by placing two dashes "—-" in columns 1 and 2. The input files use indentation (3 spaces per level) to denote a branch from a given node. An exclamation point and a space '! ' signify the starting point for indenting levels. Once defined, these exclamation points must remain in the same column for every line in the file. This is done to enhance the readability of the various levels of the tree structure. If the signal is defined in a DMC, specify the DMC and Variable names BEFORE the '! ' marker. Remember the reverse logic and override node notation described in the "Mill Startup Prompting - MSUP" section above. These are specified by placing a tilde character (~) or a pound sign (#) on to the front of the dmc variable name, respectively. An include file specification follows the indentation scheme, specifying the complete file name within brackets as follows :

```
!    << filename.filetype >>
```

Each file should be limited to approximately 2 pages of branching information, to allow the user to easily see all branches from a given node without confusion.

Nesting of include files is permitted up to a level of 100 files, at which point the array limits within this program should be increased. The maximum size of the output direct-access file is specified as 'database_size' records long, with database_size initially set to 10,000 records.

The following figure provides a sample input file specification.

```
-- Sample input file specification for CONVERT_TREE :
--
-- FAULT TREE      :  Finishing_Mill.TREE
-- WRITTEN         :  9/20/88
-- AUTHOR          :  Roy G. Graham
--
-- ABSTRACT        :  This is the top level of the Mill Start Up Prompting
--                    (MSUP) fault tree. This is the file that should be
--                    converted using the CONVERT_TREE utility.  The first
--                    branch contains only a short hardware check.  The
--                    other branches contain more detailed checks on each
--                    subsystem.
--
-- Example lines (remove the "--") :
--  DMC  ~VARNAME      ! Comment area    ( ~ means the bit should be false)
--  DMC  VARNAME       !    Indent 3 spaces for each level of the tree
--
-------------------------------------------------------------------------------

                  ! Controller Hardware  POWERED ON
                  !    <<POWERED_ON.TREE>>
                  ! Man Machine Interface system READY
                  !    <<MMI.TREE>>
                  ! Drives READY
                  !    <<MAIN_DRIVE.TREE>>
                  ! Hydraulic systems READY
                  !    <<HYDRAULICS.TREE>>
                  ! Position regulators READY
                  !    Roll Gap control
                  !       Screwdown position regulators
                  !          <<SCREWDOWN.TREE>>
                  !       <<CYLINDERS.TREE>>
                  !    Sideguides
                  !       <<SIDE_GUIDE.TREE>>
                  !    Side shifting
                  !       <<SIDE_SHIFTING.TREE>>
                  ! Coolant system ready
                  !    <<COOLING.TREE>>
                  ! Cropshear subsystem READY
                  !    <<CROPSHEAR.TREE>>
                  ! Remaining Automation systems READY
                  !    Classifier / Monitor system READY
                  !       <<CLASSIFIER.TREE>>
                  !    Roll Bending system READY
                  !       <<ROLL_BENDING.TREE>>
                  !    Holding table covers READY
                  !       <<HOLDING_TABLES.TREE>>
                  !    Run out table sprays READY
                  !       <<ROT_SPRAYS.TREE>>
```

This is a sample tree for the highest level of the permissive tree. An example of one of the "include" files specified above is given below. This example is from the <<ROLL_BENDING.TREE>> include file listed above.

```
-- FAULT TREE      :  DRB.TREE
-- WRITTEN         :  9/14/88
-- AUTHOR          :  Roy G. Graham
--
-- ABSTRACT        :  This tree determines if the roll bending subsystem is
--         functional.
--
--  DMC  ~VARNAME      ! Comment area     ( ~ means the bit should be false)
--  DMC  VARNAME       !    Indent 3 spaces for each level of the tree
--
-------------------------------------------------------------------------------
DRB  ONLINE         ! CONTROLLER ONLINE - Roll Bending DMC (DRB)
                    ! I/O HEALTHY
DRB  ~S44OFFL       !    Local I/O HEALTHY
DRB  FUSESTAT       !    I/O Power supply is HEALTHY
DRB  RBNIDOK        ! Other required controllers ONLINE
DRB  ZDM1VLN        !    DM1 ONLINE - Main Drive DMC 1
DRB  ZDM2VLN        !    DM2 ONLINE - Main Drive DMC 2
DRB  CMFONLK        !    CMF ONLINE - FM MMI MicroVAX
DRB  ZDBRVLN        !    DBR ONLINE - Roll Change S/6
                    ! Roll bending subsystem HEALTHY
DRB  ~FMLSTOP       !    Finishing mill not stopped
DRB  DM1_FMLESRK    !       E-stop has been reset
DRB  ~DM1_FMLSTPC   !       No fast stop command
DRB  ZDM1VLN        !       Main drive DMC  DM1 ONLINE
--
DRB  S7BNDRY        !    Stand 7 Roll Bending READY
DRB  ~S7RFLT        !       Stand 7 bending system  HEALTHY
DRB  ~DRB_RB7FORA   !          Stand 7 Roll Bending Feedback IN RANGE
DRB  ~DRB_RB7DFEA   !          Stand 7 Roll Bending differential force IN LIM
DRB  SR1_RB7HTYK    !       Aux Hydraulic system HEALTHY
DRB  S7KLKON        !       Kelk pressure transducers HEALTHY
DRB  ~S44OFFL       !       IO Station 44 communications HEALTHY
DRB  ONLINE         !       DSS ONLINE -  Side Shifting DMC
DRB  ZDA7VLN        !       Stand 7 Cylinder control DMC DA7 ONLINE
---
```

6.2.2 Running the utility

The Mill Startup Prompting Preprocessor has been designed for the database administrator. This is the utility which preprocesses the sequential input files, and creates a new version of the MSUP fault tree. By providing a scheme wherein the MSUP database is stored in a series of files requiring preprocessing, the database administrator has an automatic backup source for the MSUP database, should the direct-access file become corrupted.

The CONVERT_TREE utility may be run as an interactive job or in batch. A command file (CONVERT_TREE.COM) is provided for batch execution. The CONVERT_TREE utility prompts the user for the name of the highest level input file name (the default is MSUP$DIR:MSUP.TREE), the name of the direct-access output file to be created (the default is MSUP$DIR:MSUP.DATABASE), and the level of debug output (the default is 'none').

Three levels of debug output are available :

```
'None' or 'N'   =>  Only error messages are displayed.

'Files' or 'F' =>  Error messages are displayed, along with
                   the names of all include files as they
                   are opened and closed.

'All' or 'A'    =>  Error messages, file names, and every
                   record used to create the fault tree
                   are displayed.
```

In terms of timing, the program requires approximately 16 seconds to initialize a 10,000 record database when run on a DEC VAX 8530, and an additional 50 seconds to establish every 100 records while displaying full debug, versus 40 seconds with debug set to 'none'. These values are approximate, and are included to provide the database administrator with an idea of the time involved in setting up a large database (10,000 records could take 50 minutes).

6.3 MSUP Status Values

The following sections provide details about each MSUP status return value which are not provided in the Operator Interface section.

6.3.1 Normal Status Values

READY
Direct status obtained by successfully talking to a DMC. The DMC recognizes the symbol requested, and reports the node's status to be in the desired state.

NOT_READY
The DMC recognizes the symbol requested, and reports the node's status is not in the desired state.

6.3.2 Permissive Tree Errors

Permissive tree errors are generated when MSUP encounters errors in the input database (MSUP.DATABASE). Contact the "MSUP database administrator" for your site to report errors in the permissive tree.

NOT_IN_DMC
The requested symbol name cannot be found at the specified station address (DMC). Note that while this is indicative of an error in the permissives tree, it is generated by correctly communicating with the DMC.

NOT_A_LOGICAL
The requested symbol name is defined in the DMC, but is an integer or real, not a logical value. As in the NOT_IN_DMC status value, this status is generated by correctly communicating with the DMC.

BAD_LEAF_NODE
A leaf node was found without a signal name to evaluate. This is an illegal condition, since a "leaf" has no branches, and the only way to determine the status of this type of node is to directly read the DMC variable. The "database administrator" needs to add DMC variable information to this node.

BAD_BYPASS
A bypass node was found without a subtree. A bypass node is a special type of node in which traversal of a branch beneath the node can be controlled by the value of a permissive (DMC variable).

6.3.3 CSF Errors

A number of errors can be generated when MSUP is unable to successfully communicate over the Control System Freeway. These errors include :

DMH_STAT_FAIL

The BMP_SEND worked, but the return status from the DMH_STATUS variable indicates error. This error might be eliminated by using the REFRESH command to re-examine the DMC variables in question.

BMP_SEND_ERR

Error when attempting to send out a directed message using the BMP_SEND routine. Usually signifies a TIMEOUT. This timeout can be due to an overloaded CSF, or a DMC which is dead. Try the REFRESH command.

BMP_RCV_FAIL

The communication from the DMC to MSUP failed. Try the REFRESH command. (The BMP_RECEIVE routine flags this error in the Directed Message Handler routines).

BAD_PACKET_ID

The packet_id (sequence #) sent by the BMP_SEND does not match the id received after the BMP_RECEIVE command. Likely cause is a delay on the CSF channel. Try the REFRESH command.

DMC_Q_IS_FULL

The DMC is currently unable to respond to the status request due to its present level of utilization. Also referred to as 'STATION CLOGGED'. Try the REFRESH command.

TIME_OUT

The DMC did not respond to the status request within a desired timeout period. Try the REFRESH command.

STN_NOT_FOUND

DMC station not found in list of DMCs. This could be an error in the permissives tree (DMC name misspelled), or in the Control and Diagnostics station configuration file.

BAD_DMC_STATE

The DMC is not online.

NOT_ACKED

The DMC did not acknowledge the message sent to it by the BMP_SEND request. The DMC could be in a bad state (i.e. SICK or DEAD), but the master file containing DMC status values has not yet been updated (otherwise this status would have been BAD_DMC_STATE).

CHANNEL_FAIL

A failure occured when attempting to open a channel to the desired station using the BMP_OPEN routine. The directed message cannot be completed.

UNKNOWN

Unable to ascertain status of variable. Try REFRESH command. This is a "catch-all" error, and should normally not be seen.

A Installation

The MSUP system is delivered as a VAX/VMS save set which may be installed using the VAX/VMS command procedure "VMSINSTAL". A completed directory for this utility includes two executable files (.EXE), a set of permissive tree source files (.TREE), and one fixed length record direct access permissive database file (.DATABASE). One executable is for the main Mill Startup Prompting function, and one for the CONVERT_TREE utility. After transferring these files from the installation tape onto your VAX system, running the VMSINSTAL command file will not only build the executable files, but also define two logical definitions. A VAX logical definition allows a device independent name to be associated with input files which the MSUP or CONVERT_TREE programs are expecting. The logical name 'MSUP_DATABASE' needs to be defined to point to the direct access database delivered with your installation packet as follows in the following example :

```
$ DEFINE/GROUP  MSUP$DIR       <your_disk:[your_directory_name]>
$ DEFINE/GROUP  MSUP_DATABASE  MSUP$DIR:MSUP.DATABASE
```

The other logical name defined by the command file is used to tell the MSUP system where the DMC station configuration file is located. This file provides MSUP with CSF addresses to the various Distributed Micro Controllers (DMC's). The customer is prompted for the location of the CSF directory on their system, and the following commands are then performed.

```
$ DEFINE/GROUP  CSF$SYSTEM  <location of CSF directory>
$ DEFINE/GROUP  STA_CONFIG  CSF$SYSTEM:CSF_STATION_CONFIGURATION.DAT
```